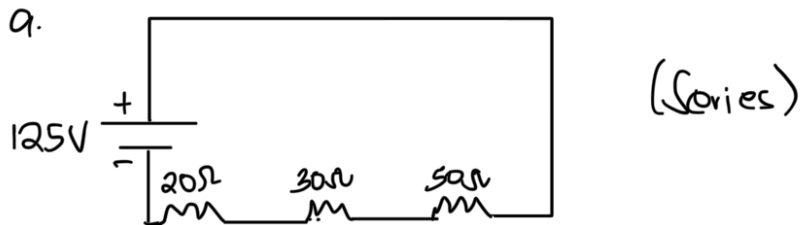


Practical Exercise 2 : Circuit Design

Part A



1. Calculate the total resistance in ohms

$$20\Omega + 30\Omega + 50\Omega = 100\Omega$$

$$\text{Ans} = 100\Omega$$

2. Calculate the total current from the power supply.

$$I = \frac{V}{R} \quad \therefore I = \frac{125}{100}$$

$$\text{Ans} = 1.25 \text{ A}$$

3. Calculate the current through each resistor.

In series current is the same through all resistors.

$$\therefore \text{Ans} = 1.25 \text{ for each resistor.}$$

4. Calculate the voltage drop through each resistor.

$$V = IR$$

$$R_1 = 1.25 \times 50 = 62.5 \text{ V}$$

$$R_2 = 1.25 \times 30 = 37.5 \text{ V}$$

$$R_3 = 1.25 \times 20 = 25 \text{ V}$$

$$\therefore 62.5 \text{ V} + 37.5 \text{ V} + 25 \text{ V} = 125 \text{ V}$$

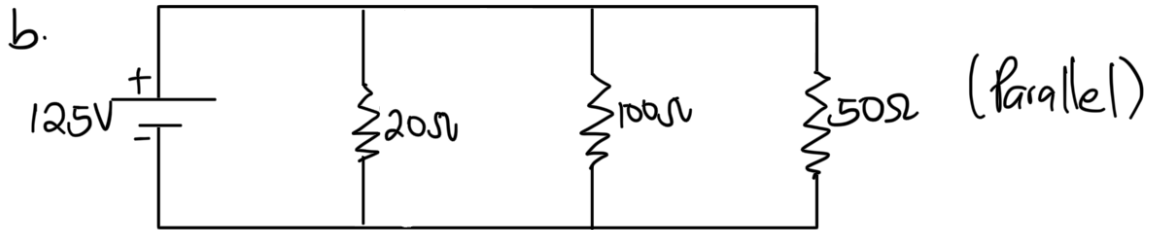
5. Calculate the power dissipated through each resistor.

(use Watt's Law : Power = VI)

$$R_1 = 62.5 \times 1.25 = 78.125 \text{ W}$$

$$R_2 = 37.5 \times 1.25 = 46.875 \text{ W}$$

$$R_3 = 25 \times 1.25 = 31.25 \text{ W}$$



1. Calculate the total R in ohms.

$$\frac{1}{\text{Total}_R} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = 0.08$$

$$= \frac{8}{100}$$

$$\therefore R = \frac{100}{8} = 12.5 \Omega$$

$$A_{ns} = 12.5 \Omega$$

2. Calculate the total current from the power supply.

$$I = V/R \quad \therefore \frac{125V}{12.5\Omega}$$

$$A_{ns} = 10A$$

3. Calculate the current through each resistor.

$$I = V/R$$

Voltage is the same across all for parallel

$$\therefore R_1 = \frac{125}{20} = 6.25 \text{ A}$$

$$R_2 = \frac{125}{100} = 1.25 \text{ A}$$

$$R_3 = \frac{125}{50} = 2.5 \text{ A}$$

$$\therefore 6.25 + 1.25 + 2.5 = 10 \text{ A}$$

4. Calculate the voltage drop across each resistor.

In parallel voltage is the same across all resistors.

$$\therefore \text{Ans} = 125 \text{ V}$$

5. Calculate the power dissipated through each resistor.

$$P = VI$$

$$R_1 = 125 \times 6.25 = 781.25 \text{ W}$$

$$R_2 = 125 \times 1.25 = 156.25 \text{ W}$$

$$R_3 = 125 \times 2.5 = 312.5 \text{ W}$$